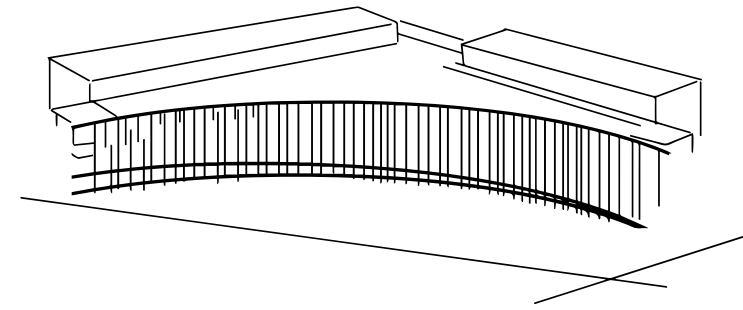




ARIES: Reciprocal-weighted Similarity of Language Model Embeddings for Multiple Sequence Alignment

Minh Hoang, Isabel Armour-Garb, Mona Singh
Lewis-Sigler Institute of Integrative Genomics
Princeton University



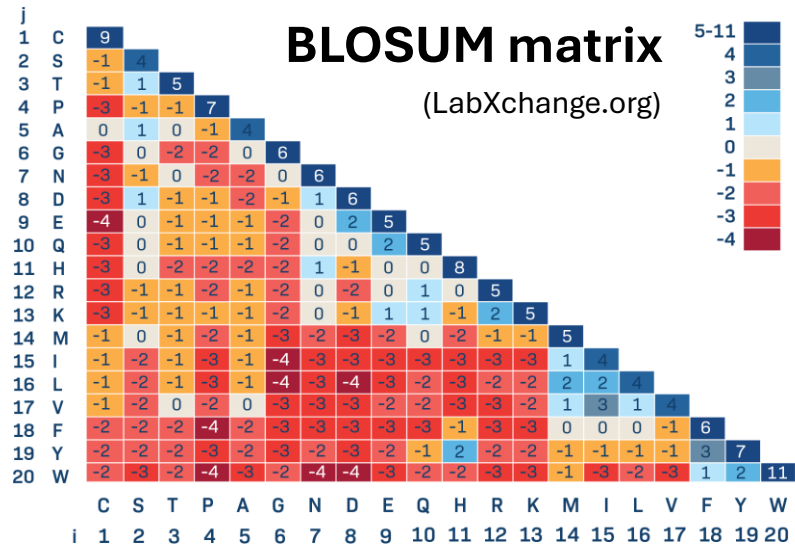
Isabel



Mona

[illegible]

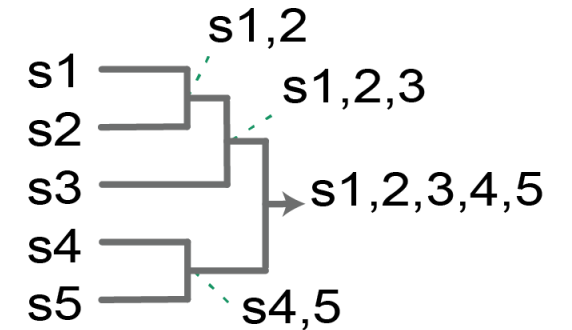
Most MSA tools rely on empirically derived substitution matrix



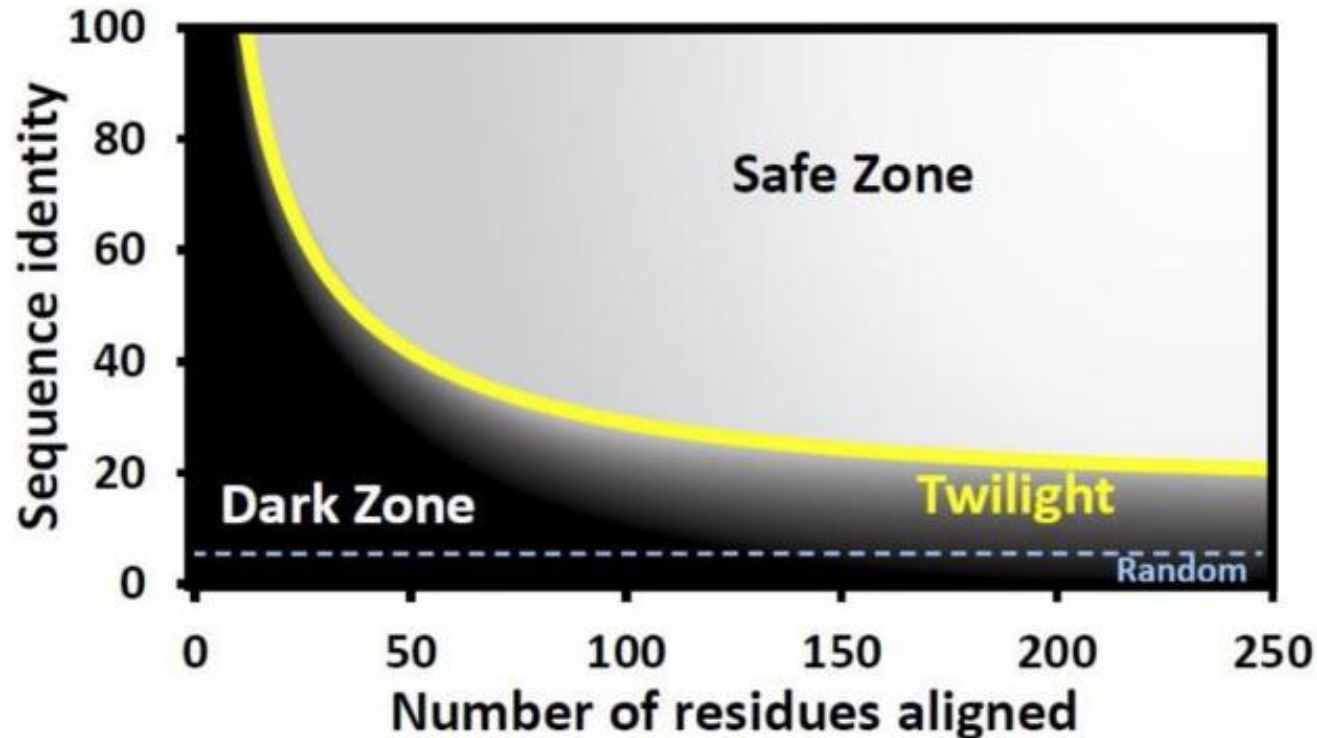
Pairwise Alignment



Progressive MSA

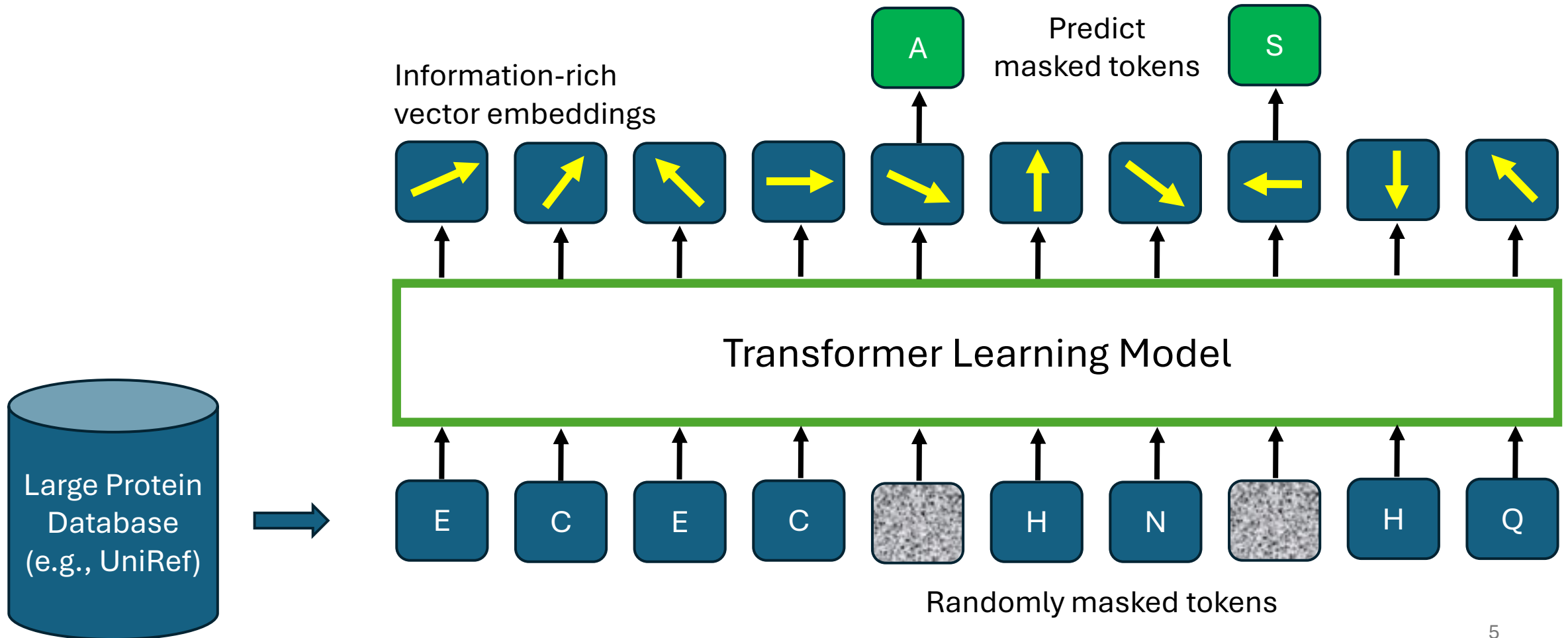


Context-free similarity scores provide poor guidance in twilight zone

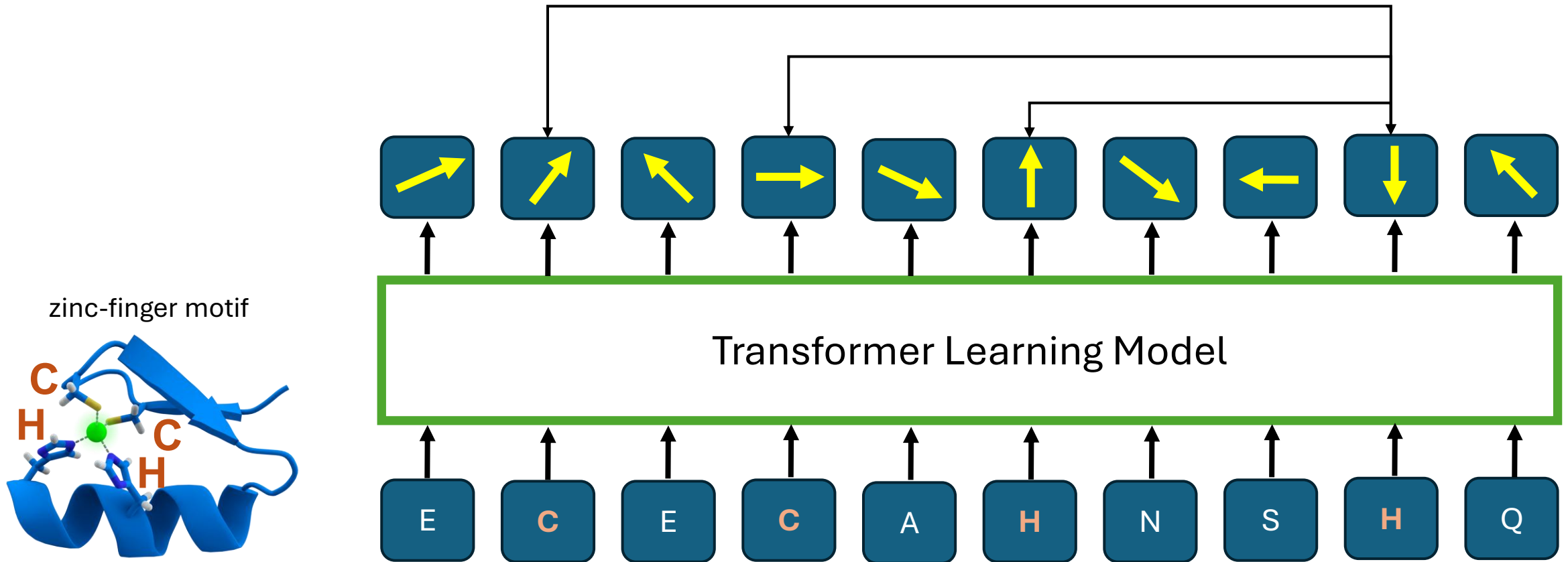


- Static substitution matrices assign context-blind scores → poor proxy for homology when sequence id falls below ~30%
- **MSA accuracy degrades precisely where it matters most:** divergent protein families, remote homologs, novel folds.

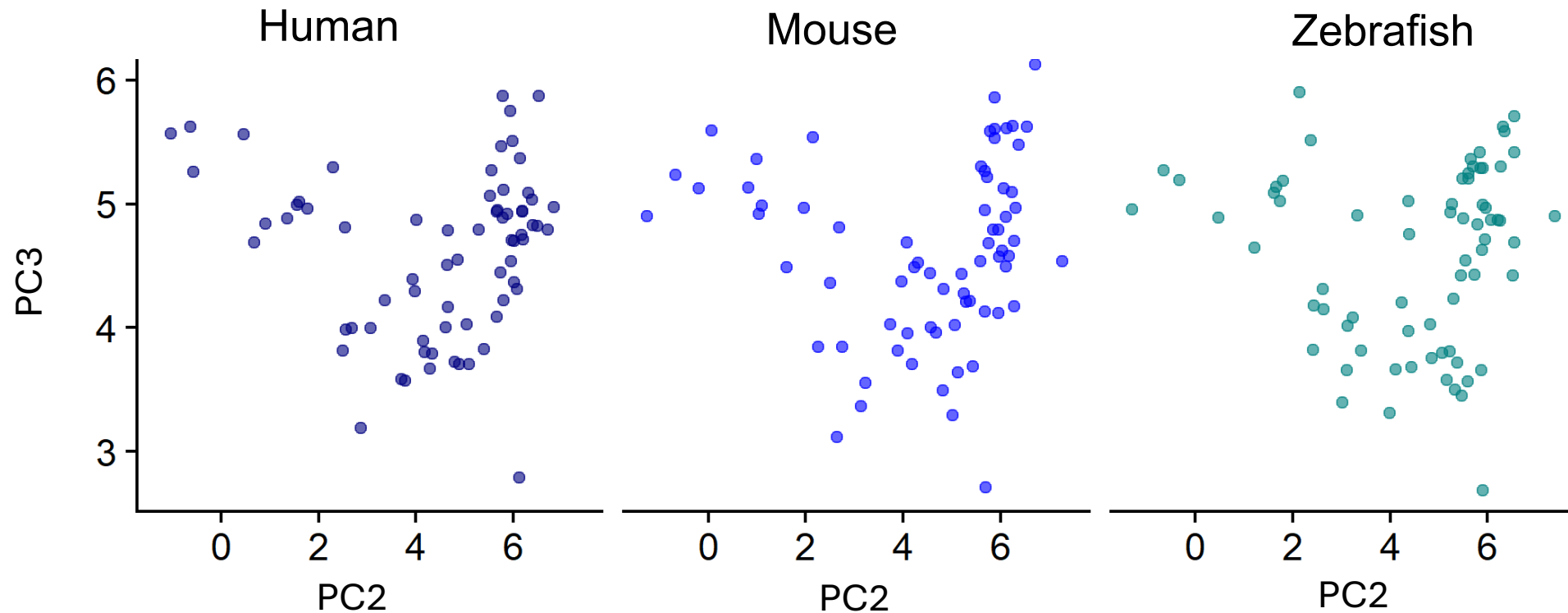
Protein language models learn AA relationship at distributional level



Protein language models learn AA relationship at distributional level

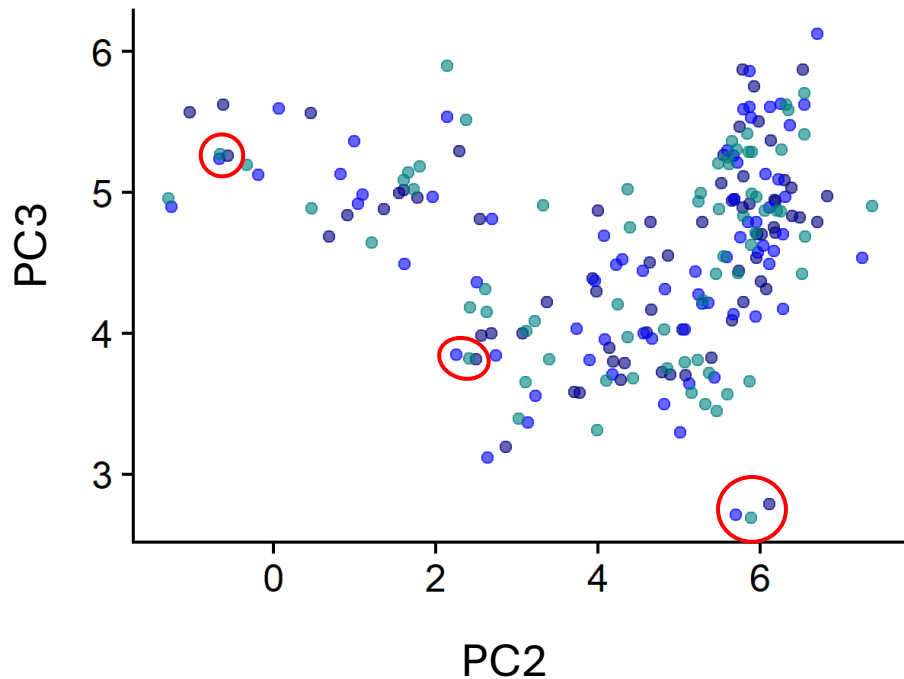


Intuition: **homologous proteins** from different species have **similar amino acid embeddings**



* Each point = amino acid representation, projected into two dimensions

Overlaying **homologous proteins** from multiple species suggests a new MSA strategy



- Each point = AA representation, projected into two dimensions.
- Each color = different species.

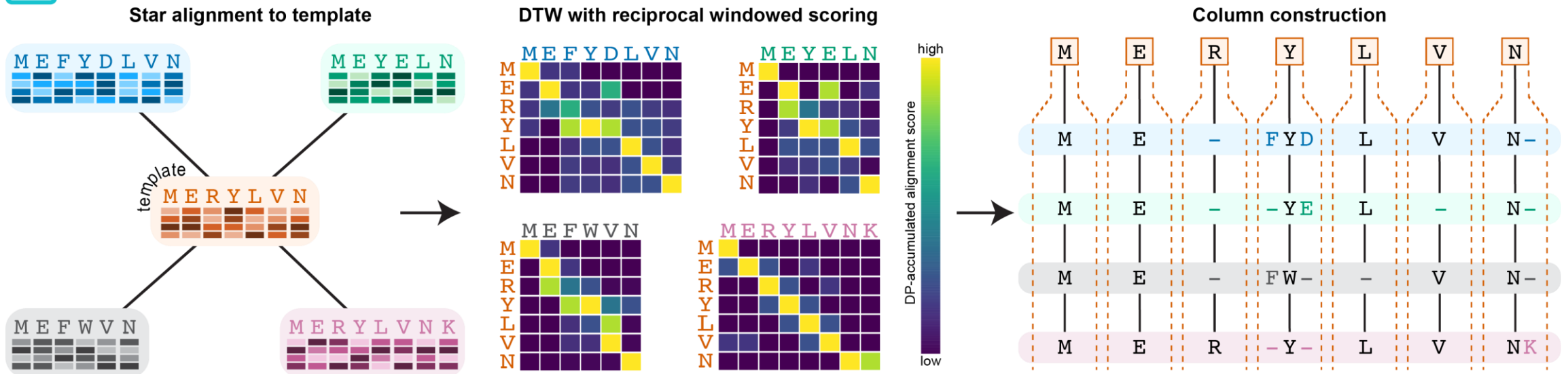
- Similar amino acid representations correspond to “analogous” amino acid positions across sequences
- vcMSA: Vector-clustering multiple sequence alignment (McWhite, Armour-Garb and Singh, 2023)



ARIES (Alignment via Reciprocal Embedding Similarity)

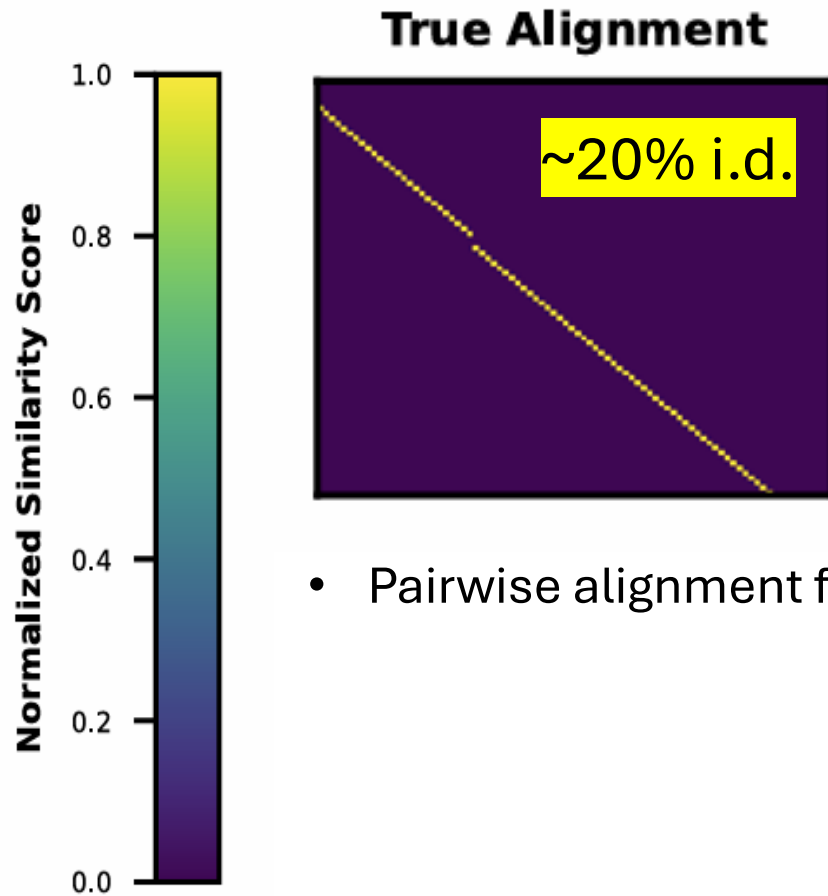


ARIES alignment module



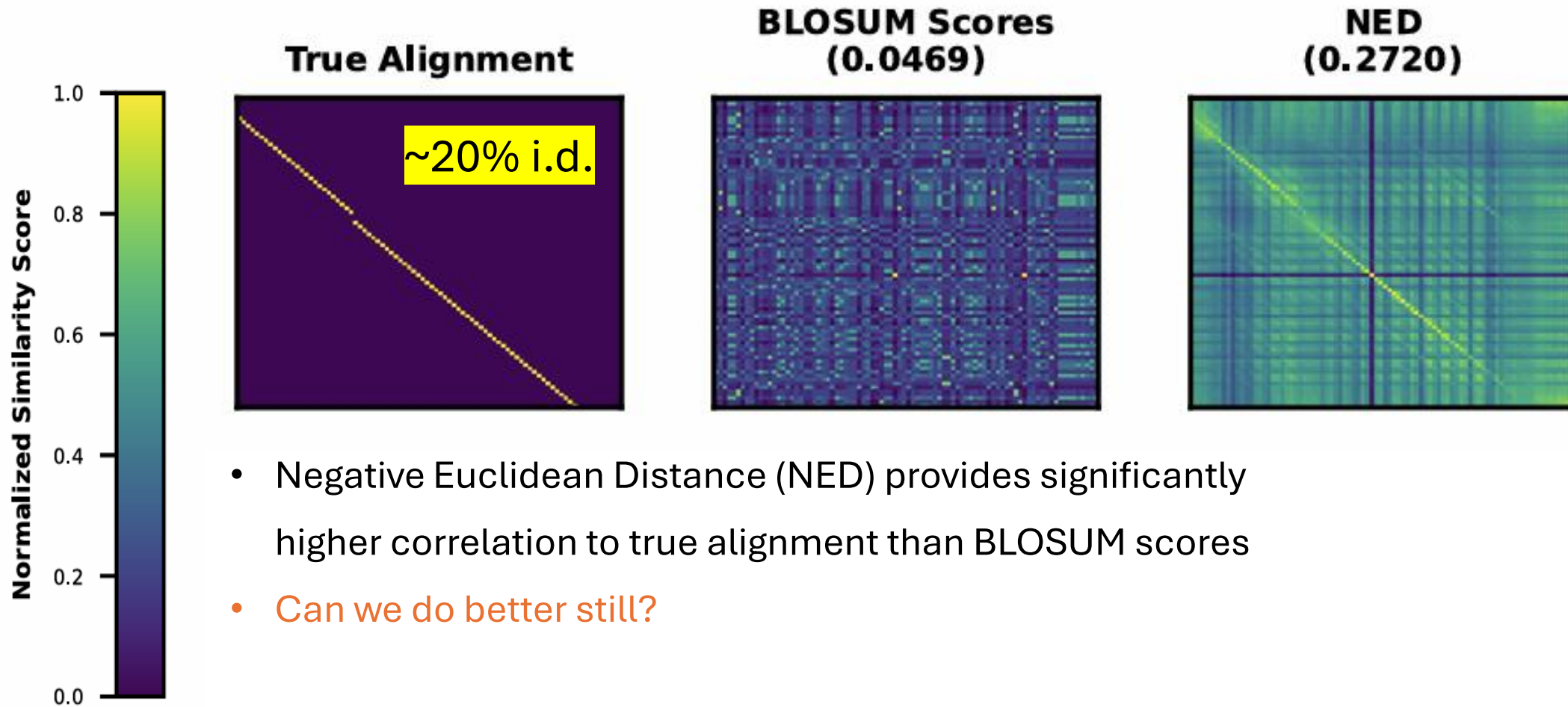
- Reciprocal embedding similarity
- Dynamic Time Warping sidesteps the need for gap penalty
- Star alignment + Template construction

Embedding similarities as position-specific substitution matrices



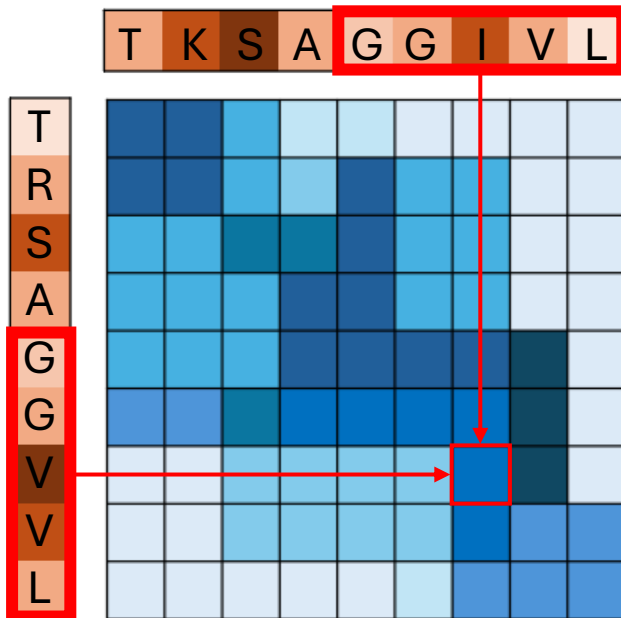
- Pairwise alignment from BALiBASE RV11 with low id

Embedding similarities as position-specific substitution matrices



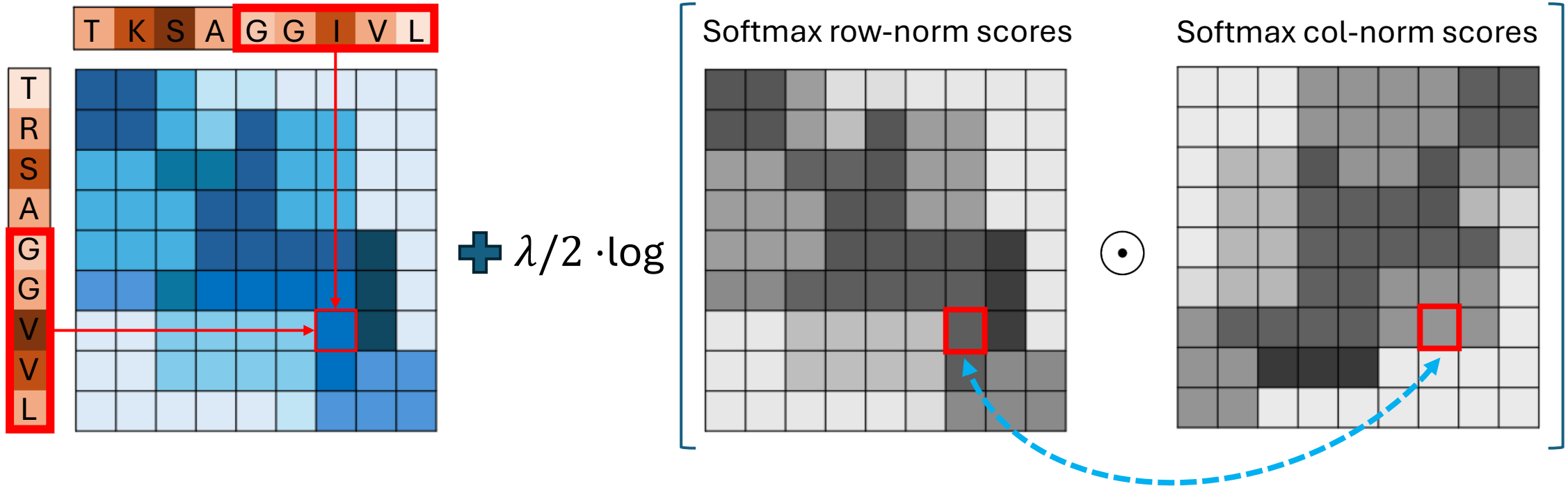
Reciprocal windowed similarity enhances alignment detection

**Window-based similarity
incorporates local context**



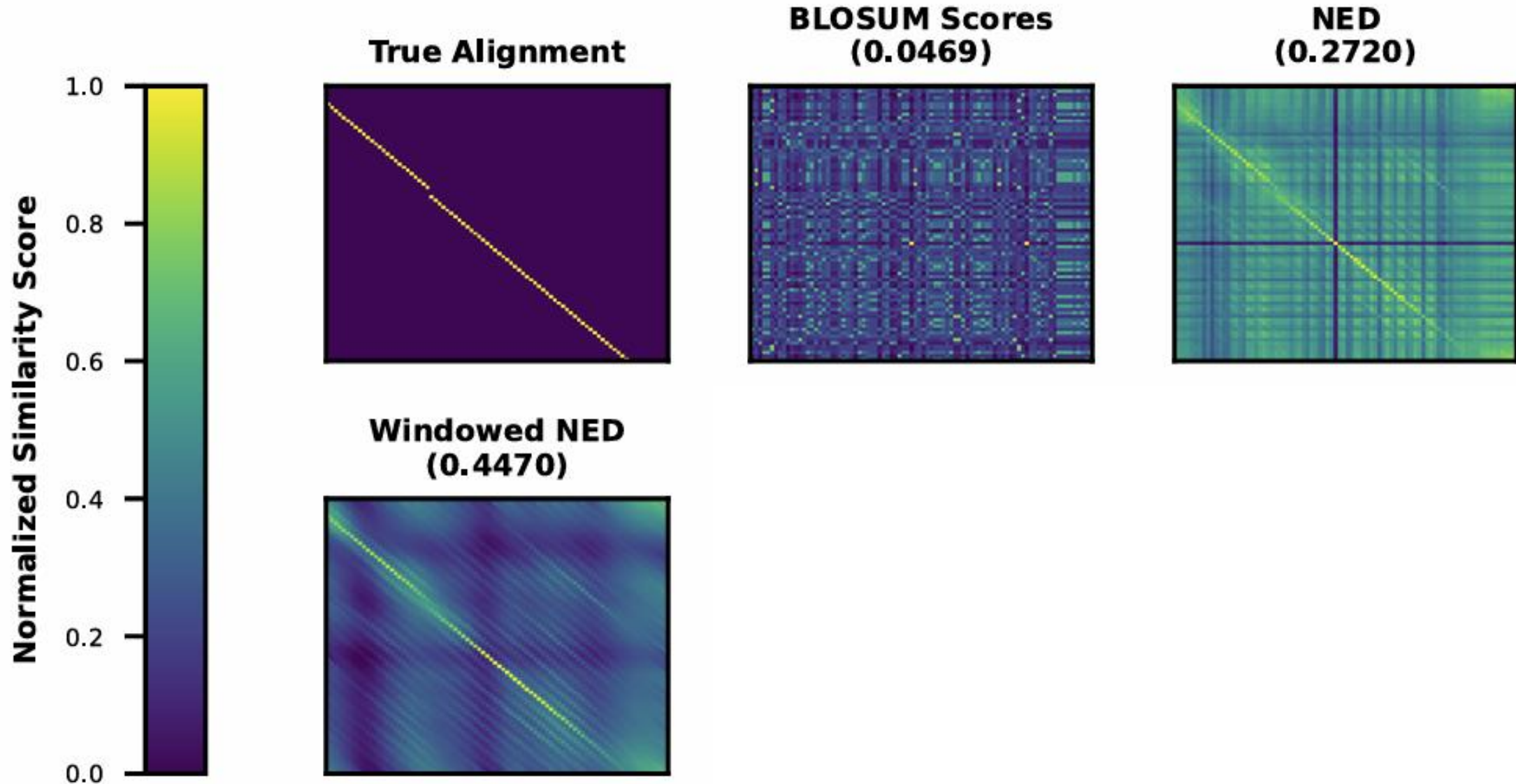
Reciprocal windowed similarity enhances alignment detection

**Geometric mean captures
mutual preference**

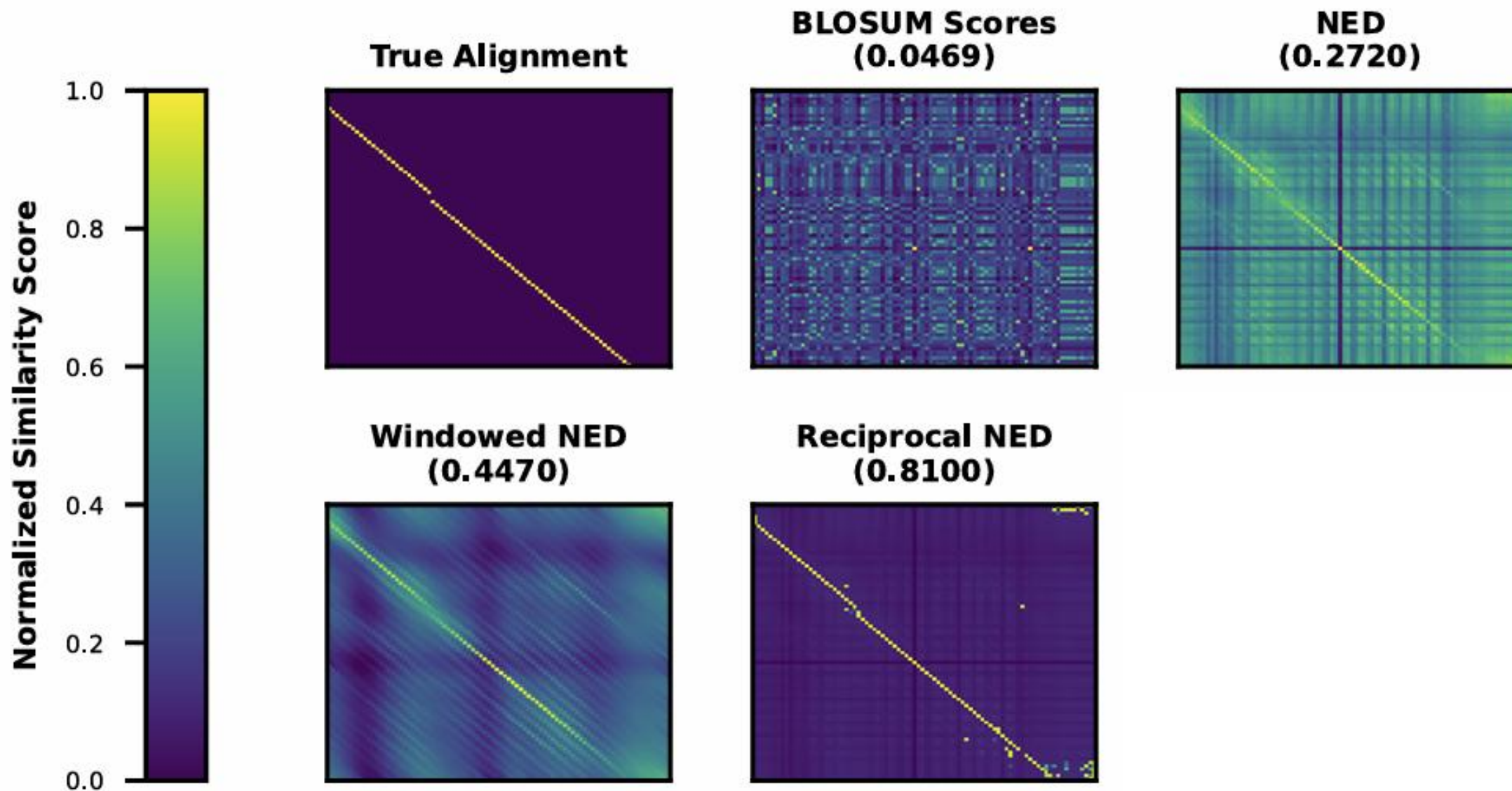


Reciprocal preference when both are high

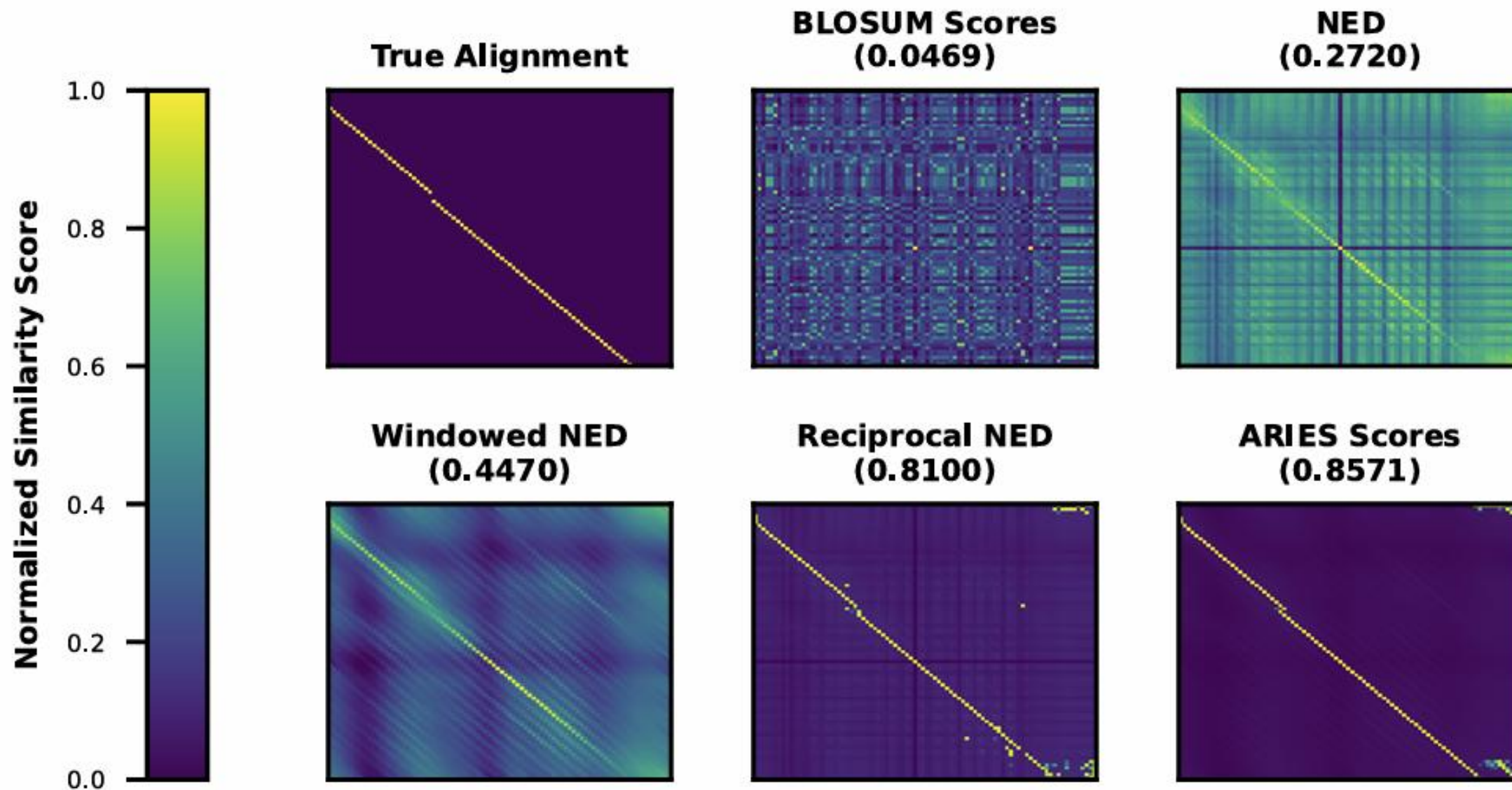
ARIES scores provide strong pairwise alignment signal



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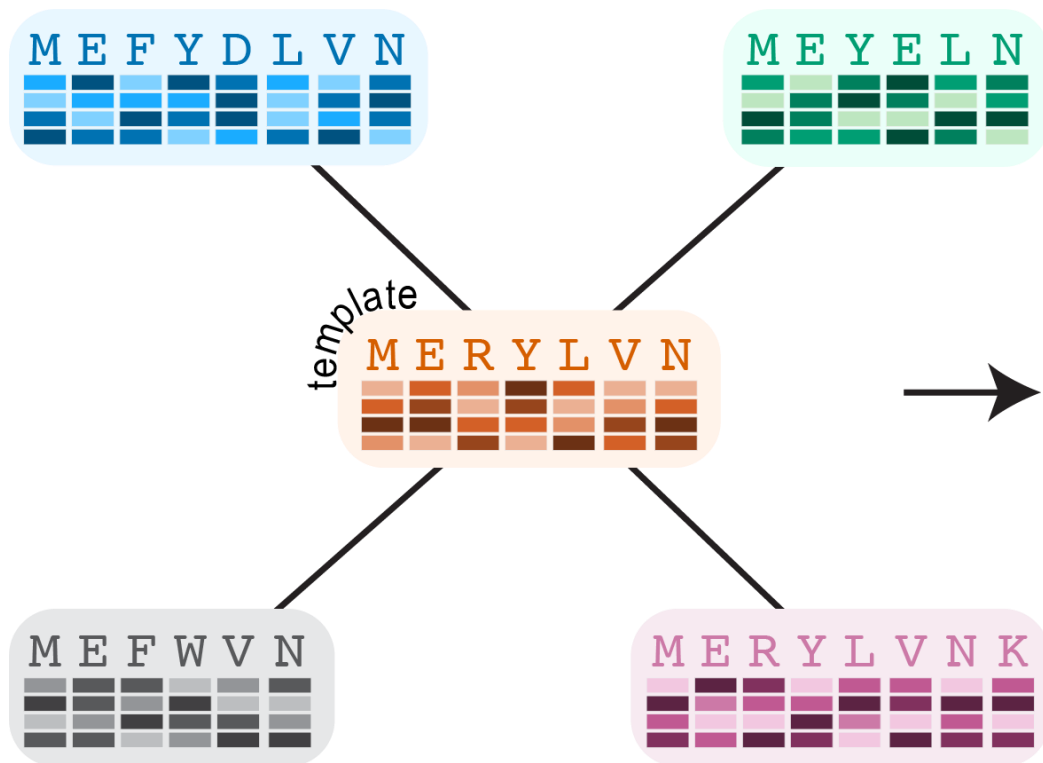


Pairwise alignment via Dynamic Time Warping (DTW)

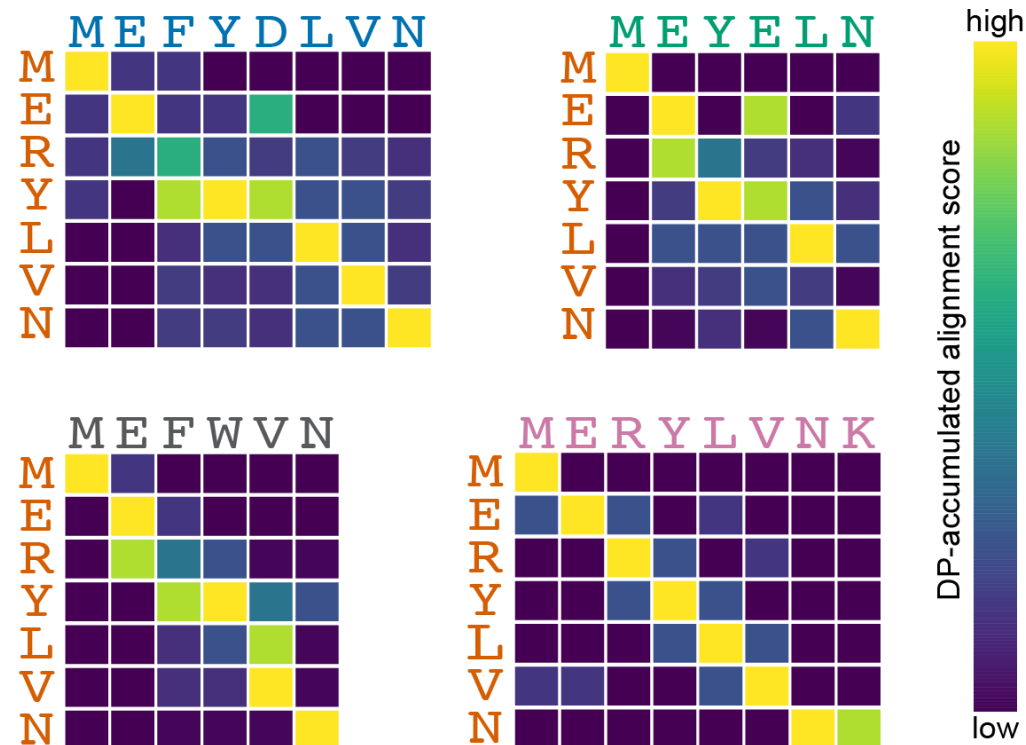


ARIES alignment module

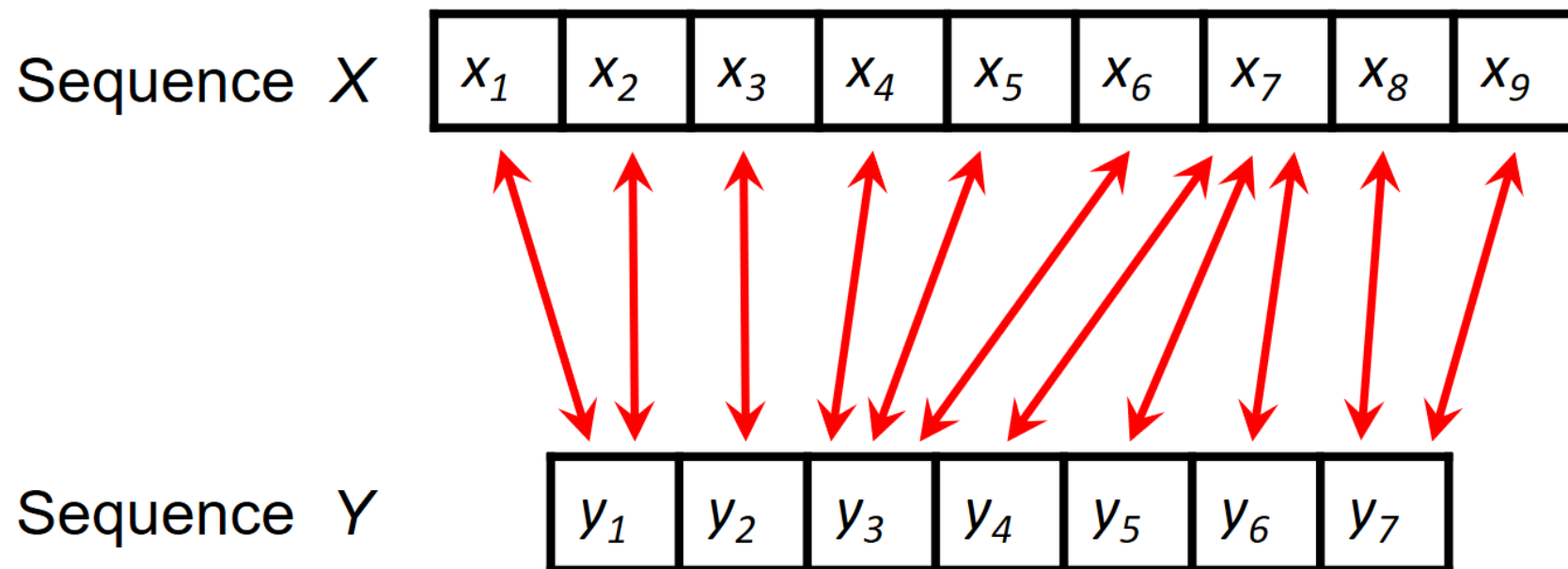
Star alignment to template



DTW with reciprocal windowed scoring



Pairwise alignment via Dynamic Time Warping (DTW)



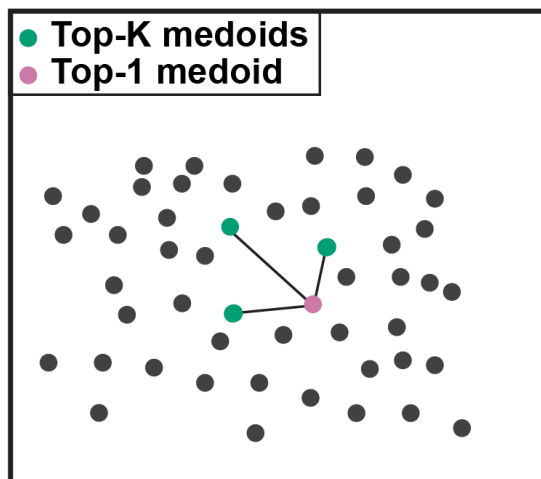
- DTW computes gap-free pairwise alignment but admits **multi-to-one residue alignments**.
- Resolve these ambiguities using confidence-based assignment during MSA building.

Synthesizing template sequence from top-K medoids



ARIES template synthesis module

Medoid selection
from guide tree



Alignment
module

Align medoids
to top-1 template

M E R Y L V X

M X R Y L X X

M E X Y V V X

M E R Y L X N

Add X tokens
in gap positions

First pass:

- Obtain guide-tree by running mBed algorithm; perform DP to select top-K medoids.
- Perform **ARIES** star alignment on these medoids to form crude alignment.

Synthesizing template sequence from top-K medoids

Align medoids
to top-1 template

M E R Y L V X

M X R Y L X X

M E X Y V V X

M E R Y L X N

Add X tokens
in gap positions

Protein
Language
Model

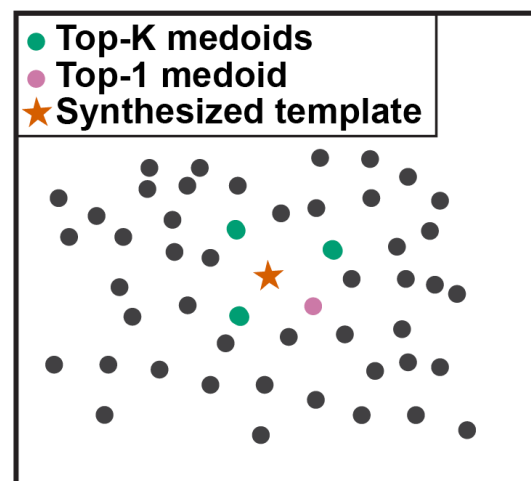
Embeddings of
aligned medoids



average



Final synthesized template
for star alignment



Second pass:

- Replace gaps in aligned medoids with 'X' and re-embed; average to form template embedding.
- Perform **ARIES** star alignment on all sequences, centered around synthesized template.

ARIES is evaluated on diverse datasets

Datasets:

- **HOMSTRAD:** 571 families whose alignments are based on structural superpositions.
- **BALiBASE 3.0:** 218 reference alignments targeting challenging scenarios such as low-identity, divergent outliers, remote sub-families, large terminal or internal extensions.
- **QuanTest 2:** 151 alignment sets with 1000 sequences, including 3 **HOMSTRAD** reference sequences, and 997 Pfam background sequences.

ARIES is compared against SotA tools

Classical baselines:

- Clustal (W, Omega)
- MAFFT (Auto, G-INS-i, L-INS-i)
- MUSCLE
- ProbCons
- T-Coffee
- FAMSA

Other non-PLM-based MSAs:

- MAGUS (Smirnov and Warnow, 2021)
- TWILIGHT (Tseng et al., 2025)

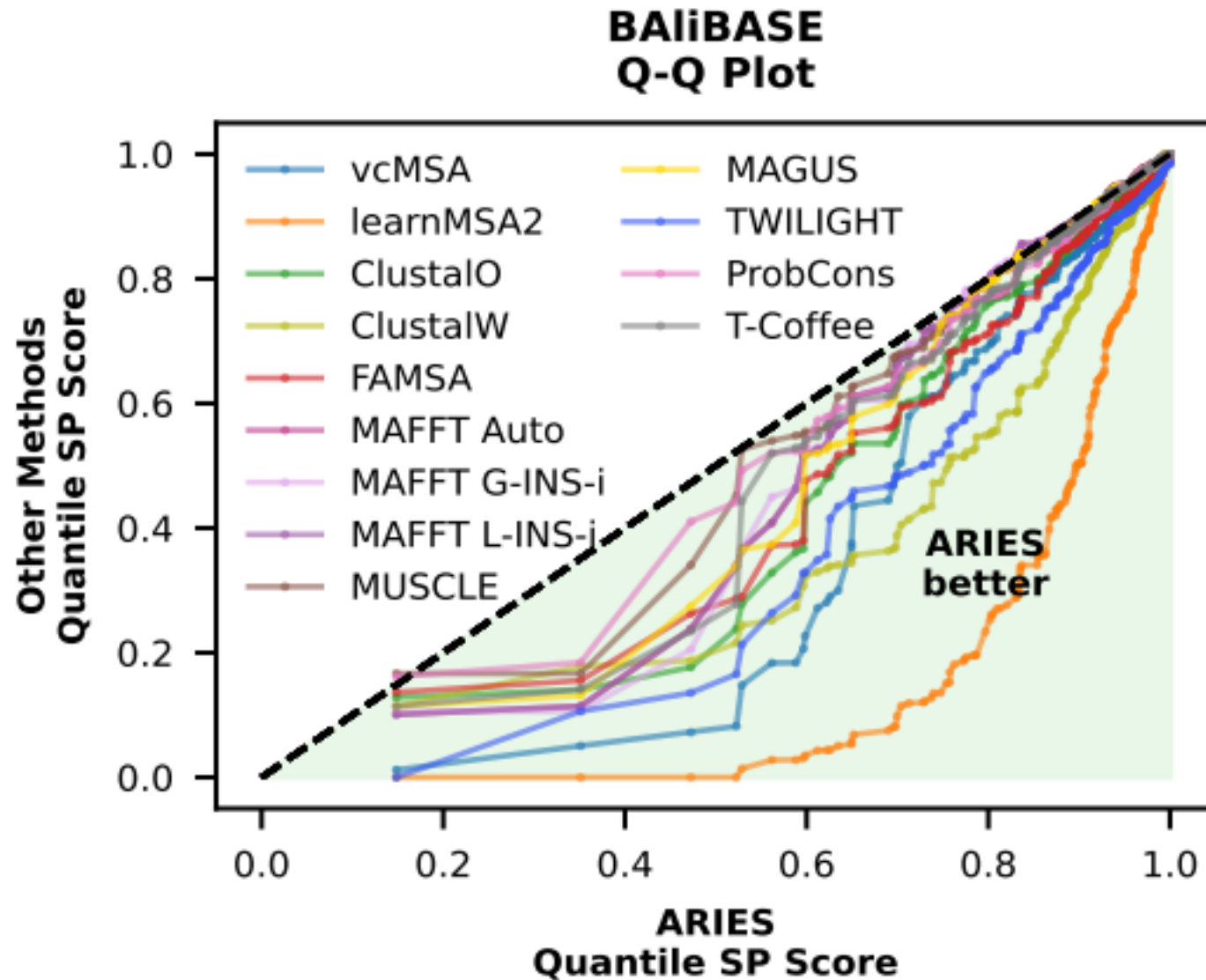
PLM-based MSAs:

- vcMSA (McWhite et al., 2023)
- learnMSA2 (Becker et al., 2024)

Metrics:

- Sum-of-Pair (SP) score: fraction of aligned pairs in gold-standard alignment.
- Total Column (TC) score: fraction of aligned pairs in gold-standard alignment.

ARIES outperforms existing methods



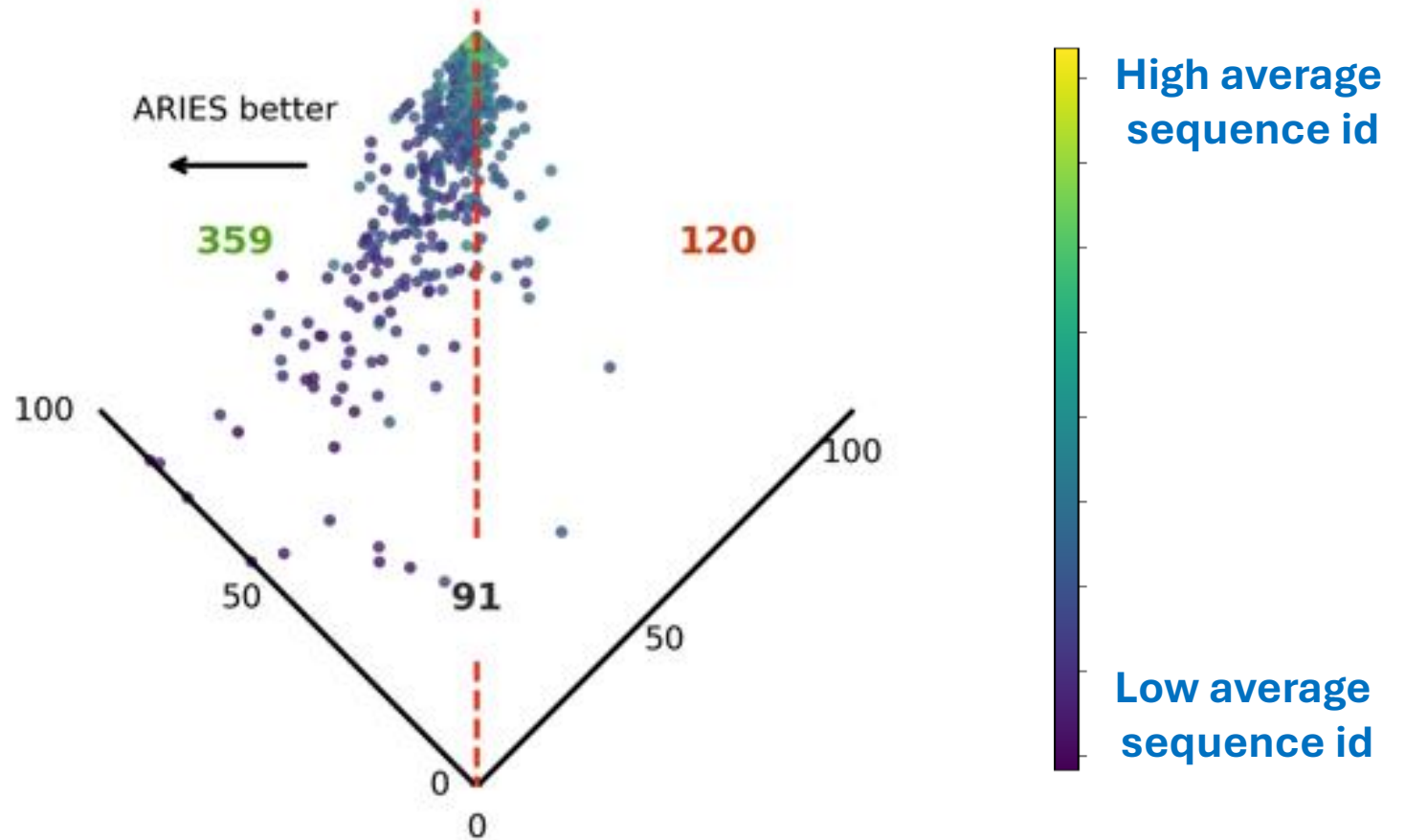
Similar performance at top quantiles (easy cases)

Big gap at lower quantiles (twilight zone)

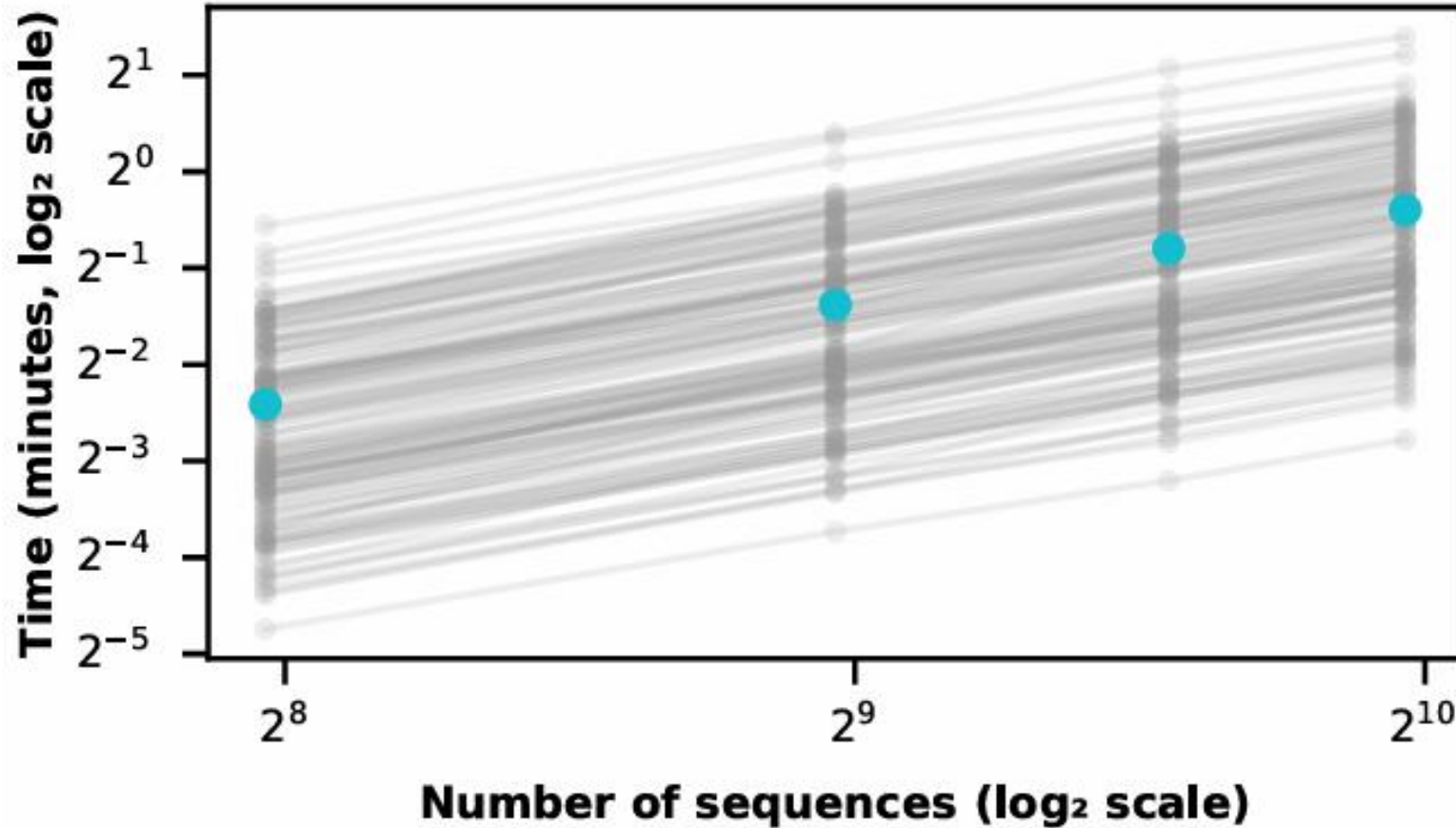
ARIES achieves better performance on low id regime

- Volcano plot shows head-to-head performance per HOMSTRAD family with next best method (MAFFT G-INS-i)
- Light: families with high average sequence identity, mostly hover around median (similar perf)
- Dark: families with low avg sequence identity (twilight zone), **heavily skewed to the left of median (ARIES wins)**

ARIES (0.90) vs. MAFFT G-INS-i (0.83)
SP Score, Wilcoxon $p = 5.761e-39$ | $n=570$

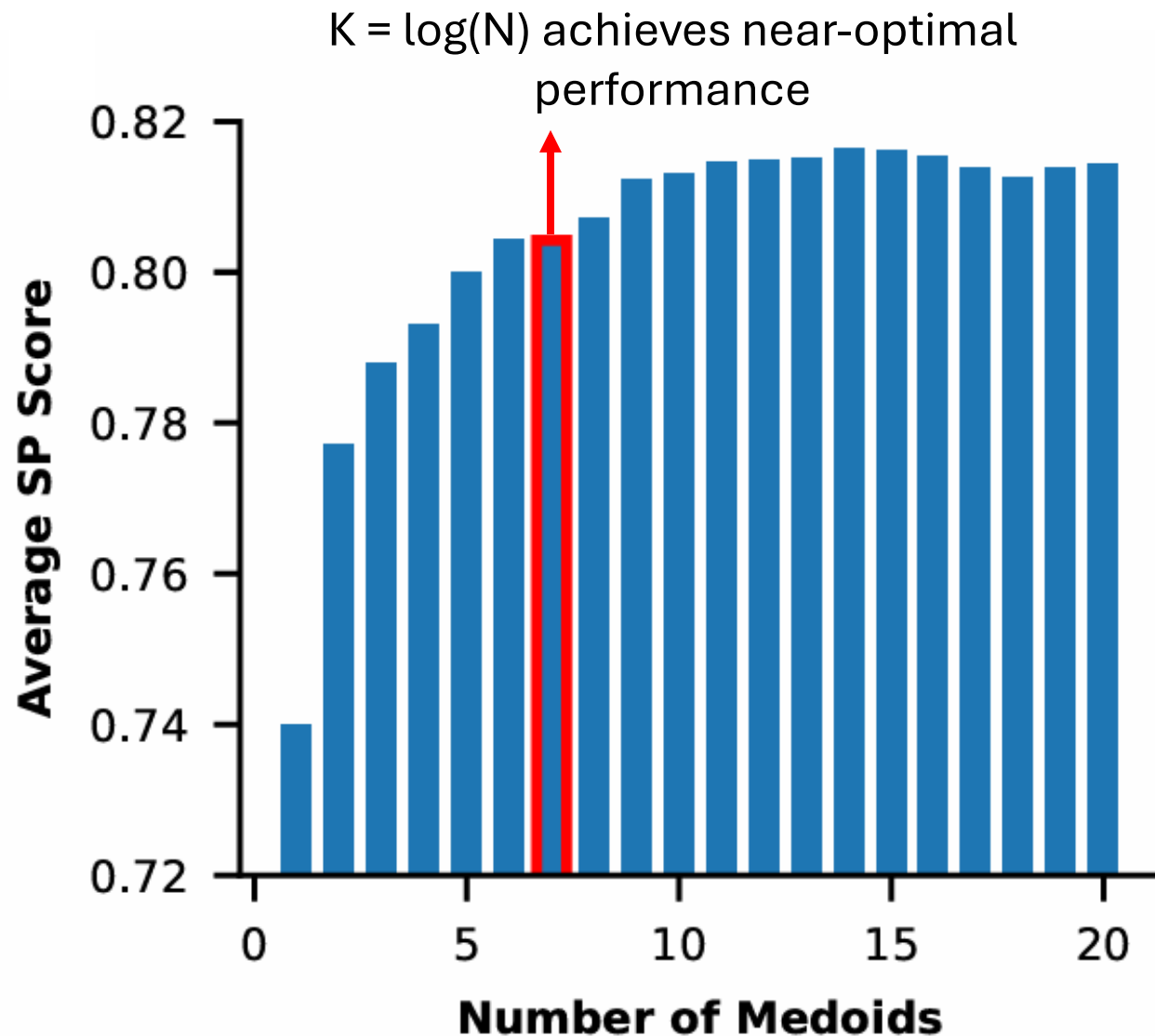


ARIES scales linearly in the number of sequences



- Each line shows the log runtime of one QuanTest family, subsampled to 250/500/750/1000 sequences.
- Blue dot indicates mean runtime across all families.
- ARIES is **5x faster** than next best PLM-based method (learnMSA2).

Template quality improves with more medoids



- Average SP score across all QuanTest families using different number of medoids to construct star template.
- Single medoid has worst performance, as expected
- Diminishing return, red bar shows **$K = \log(N)$ achieves near optimal score.**

ARIES: Harnessing PLM embeddings for efficient and accurate MSA

Summary:

- PLM-based similarity scores superior to substitution matrices
- Outperforms traditional MSA approaches, especially in “twilight zone”.
- ARIES is scalable and accurate across families of different sizes and levels of similarity.

Future works:

- Incorporating decades of algorithmic innovations into ARIES. Iterative refinement or consistency-based scoring could in principle be layered on top of an embedding-based scoring signal.
- Differentiable alignment allows fine-tuning ARIES scores to better reflect homology, especially in the twilight zone.

Check out ARIES on
our GitHub repository!



Thanks for listening! Happy to take questions

ARIES co-authors:



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Mona Singh

This research was supported in part by:



Singh Lab members:



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James Fife
Jaden Luo
Masha Larina
Sohit Miglani
Sam Neff
Nina Sachdev
Megan Schmidt
Md Toki Tahmid
Yuqi Zhang
Dr. Josh Wetzel
Dr. Yingying Zhang

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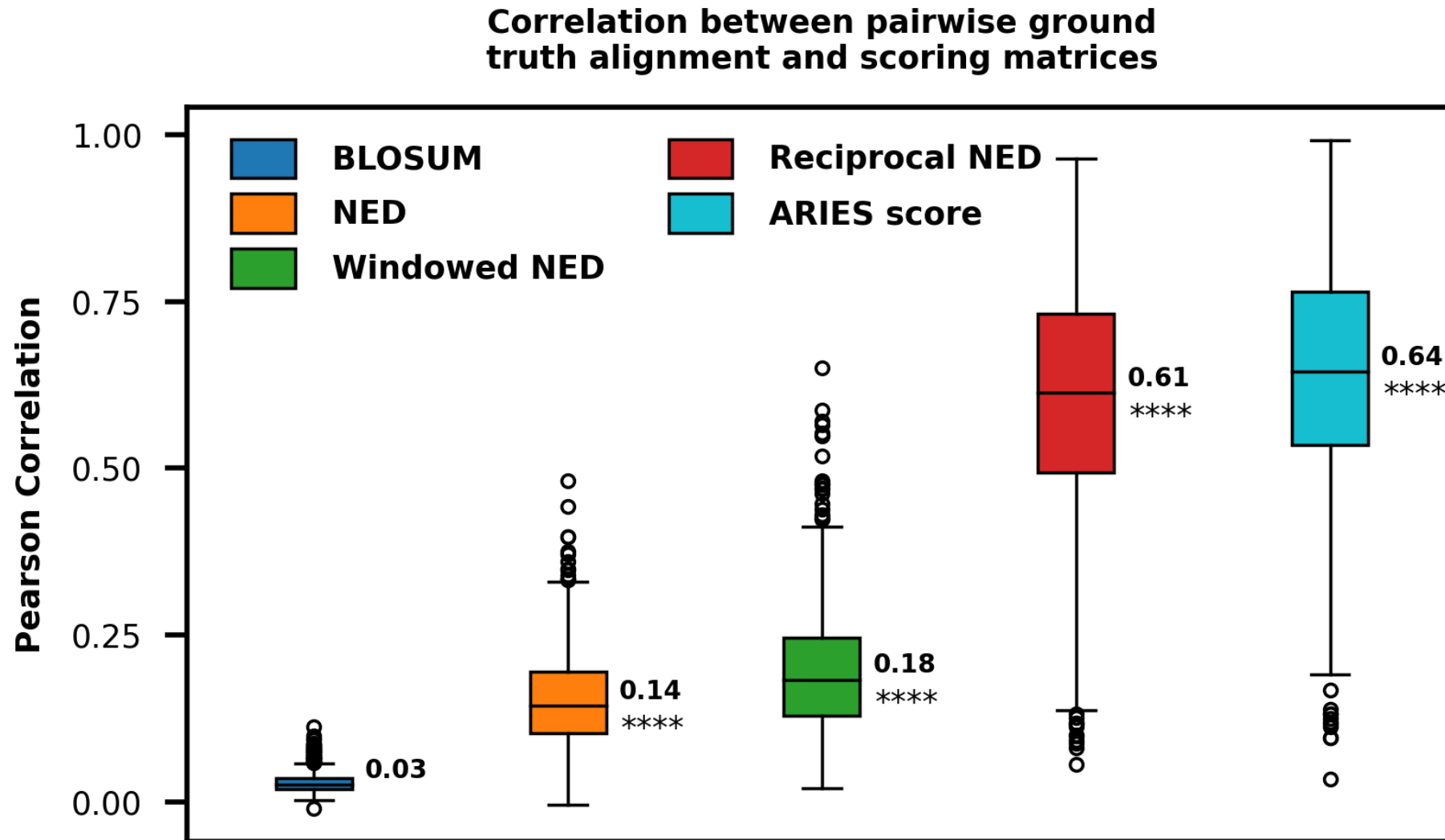
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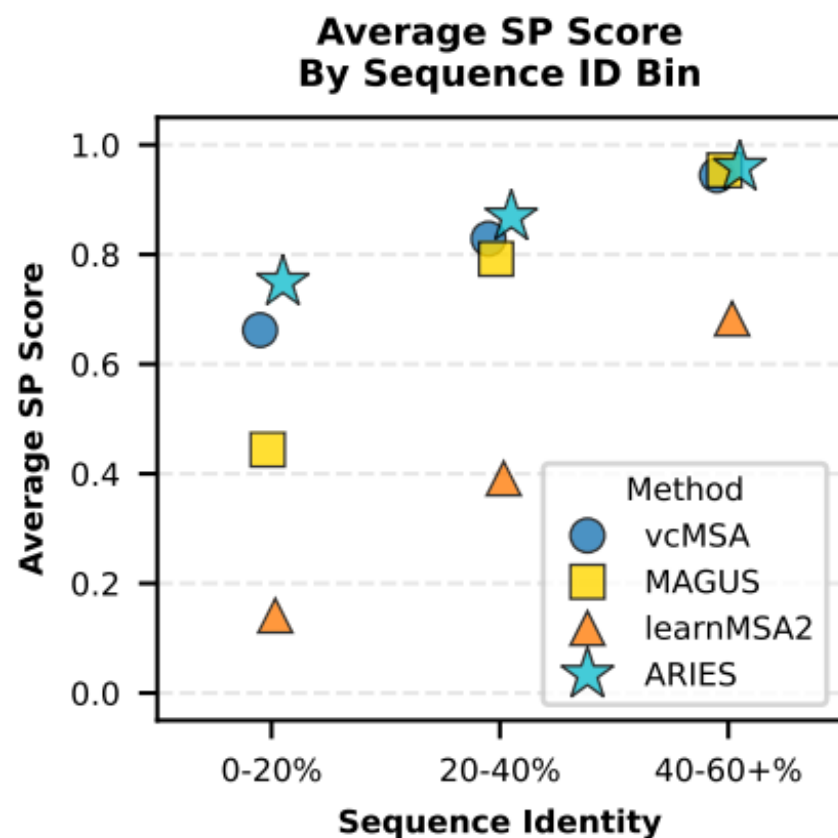


ARIES scores provide strong pairwise alignment signal

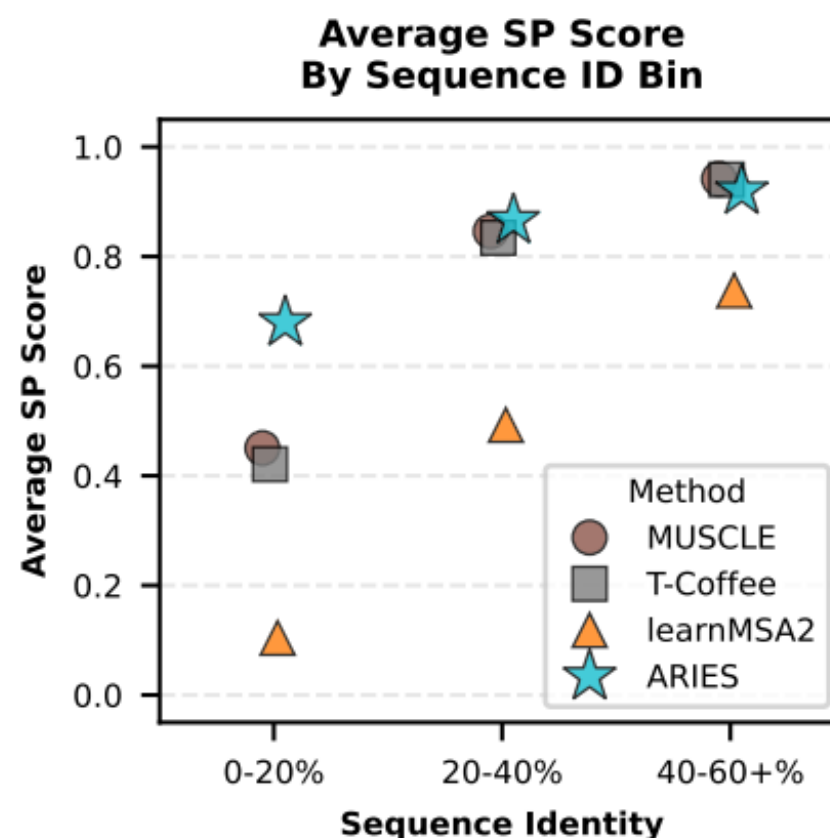


ARIES excels in low-identity MSA sets (twilight zone)

HOMSTRAD



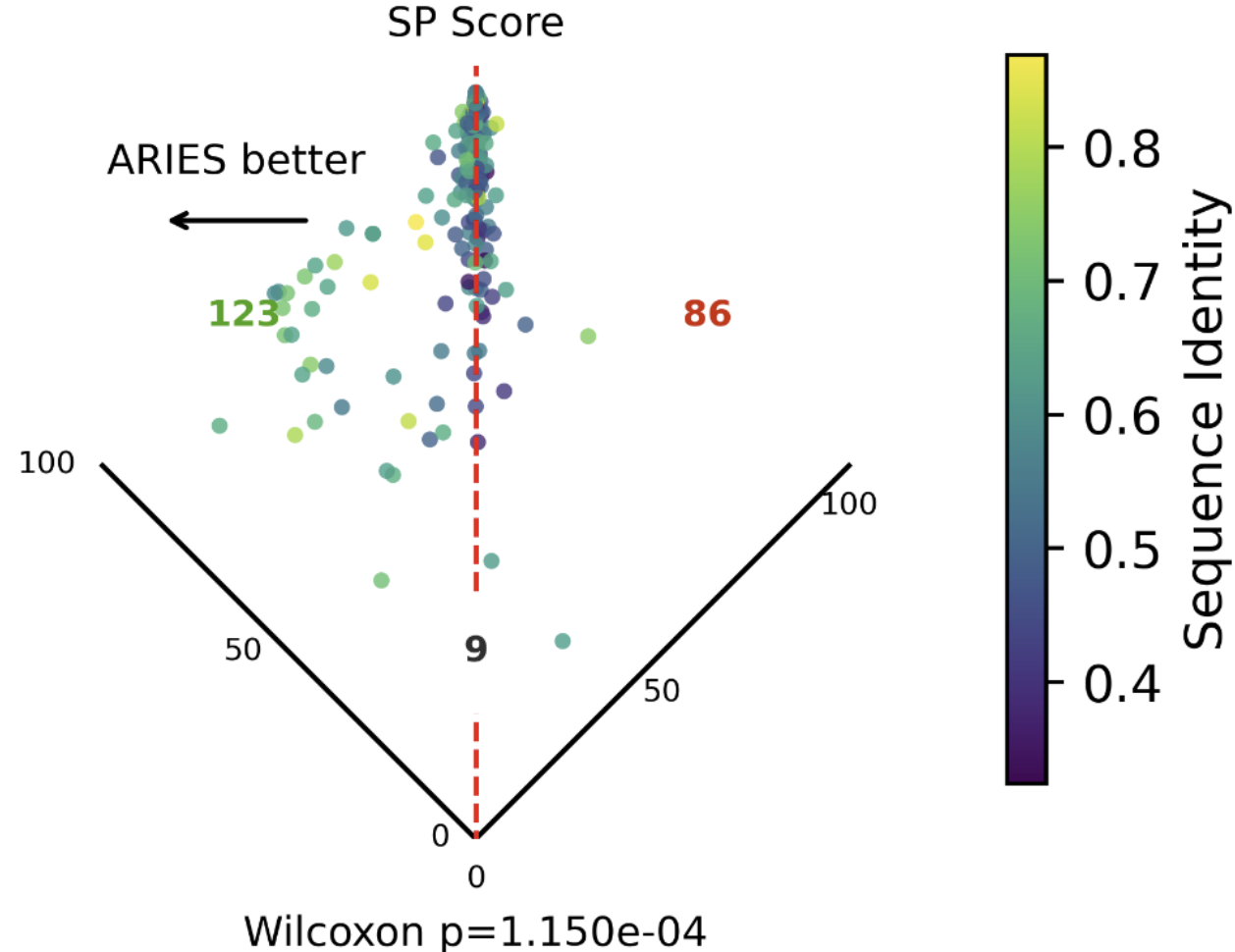
BALIiBASE



- We compared ARIES to next-best, median, and worst performing baselines per bucket of sequence identity.
- Most significant gains come from the 0-20% bucket (twilight zone).

Star-alignment vs. Progressive alignment on the embedding space

ARIES (0.88) vs. Prog. ARIES Mean-Merge (0.83)



- Progressive alignment requires an embedding-level representation for each intermediate aligned cluster before proceeding to the next merge
- These consensus comes from aggregating aligned sequences in partial profile. Signals seem to degrade with each aggregation.